

# Data Structures Lab – Assignment 5

February 18, 2026

## Assignment: Doubly and Circular Linked Lists Implementation

### Objective

To understand the structure and working of a **Doubly Linked List** and **Circular Linked List** by implementing common linked list operations such as creation, traversal, insertion, deletion, and searching. The lab aims to help students linked-list operation for a given on to be used.

### Q1: Insert at a Given Position (Doubly Linked List)

Write a program to insert a new node with a given value at position  $k$  in a doubly linked list.

#### Sample Input

Initial List: 10 20 30 40

Value to insert: 25

Position: 3

#### Sample Output

Updated List: 10 20 25 30 40

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### Q2: Reverse a Doubly Linked List

Given the head of a doubly linked list, reverse the list in-place and print the reversed list.

#### Sample Input

Initial List: 5 10 15 20

## Sample Output

Reversed List: 20 15 10 5

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## Q3: Delete All Occurrences of a Key (Doubly Linked List)

Given a doubly linked list and an integer **key**, delete all nodes containing the key.

## Sample Input

Initial List: 4 7 4 9 4 11

Key to delete: 4

## Sample Output

Updated List: 7 9 11

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## Q4: Delete the Last Node of a Circular Linked List

Create a circular linked list and delete the last node.

### Sample Input:

Initial List: 10 20 30 40

### Sample Output:

Updated List: 10 20 30

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## Q5: Count Number of Nodes

Write a program to count the total number of nodes in the circular linked list.

### Sample Input:

List: 1 2 3 4 5

### Sample Output:

Total Nodes: 5

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